



Punjab University College
of
Information Technology
“PUCIT”

Chapter 04 - Input and Output

GE-161 Introduction to Information and Communication
Technologies

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Learning Objectives

1. Explain the purpose of a computer keyboard and the types of keyboards widely used today.
2. List several different pointing devices and describe their functions.
3. Describe the purposes of scanners and readers and list some types of scanners and readers in use today.
4. Explain what digital cameras are and how they are used today.
5. Understand the devices that can be used for audio input.
6. Describe the characteristics of a display device and explain some of the technologies used to display images.
7. List several types of printers and explain their function.
8. Identify the hardware devices typically used for audio output.

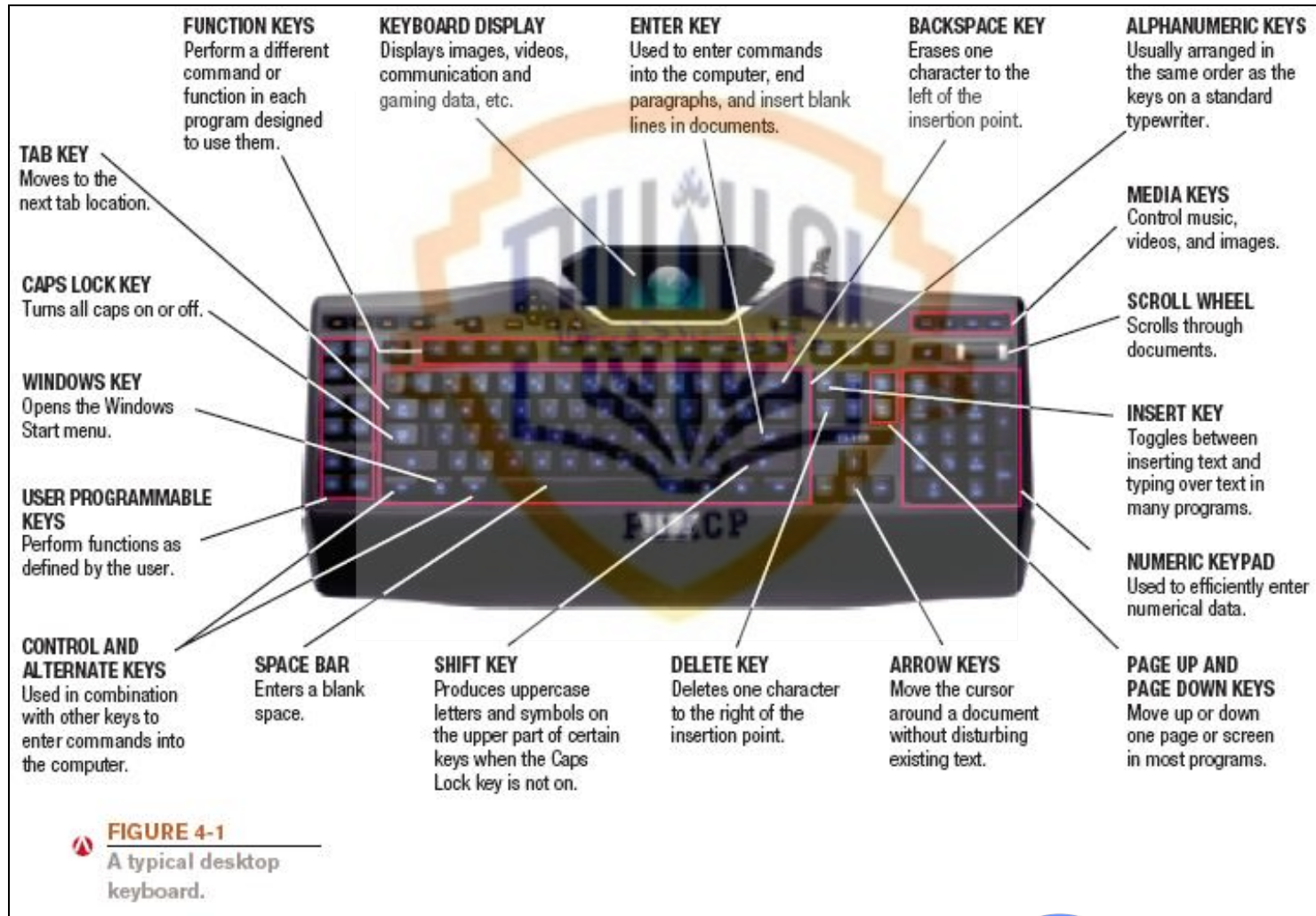
Overview

- This chapter covers:
 - Different types of keyboards and pointing devices
 - Types of scanners, readers, and digital cameras
 - Audio input devices
 - Types of display devices and how they work
 - Types of printers and how they work
 - Audio output

Keyboards

- Keyboard: An input device used to enter characters at the location marked by the insertion point or cursor
 - Can be wired or wireless
- Most computers today are designed to be used with a keyboard
- Typically contains:
 - Standard alphanumeric keys
 - Numeric keypad
 - Function keys
 - Delete and Backspace keys
 - Control and Alternate keys
 - Arrow directional keys and special keys

Keyboards



Keyboards

- Portable computers and mobile devices often use:
 - Built in or slide-out keyboard
 - Pen or touch input (on-screen keyboard)
 - Voice input



Pointing Devices

- Pointing devices: Used to select and manipulate objects
 - Used to input data
 - Used to issue commands to the computer
- Common types of pointing devices:
 - Mouse
 - Pen/stylus
 - Touch screen



Mice

- Mouse: A common pointing device that the user slides along a flat surface to move a pointer around the screen and clicks its buttons to make selections
 - Older mechanical mice use a ball
 - Optical or laser mice track with light
 - 3D mice
 - Can be wireless

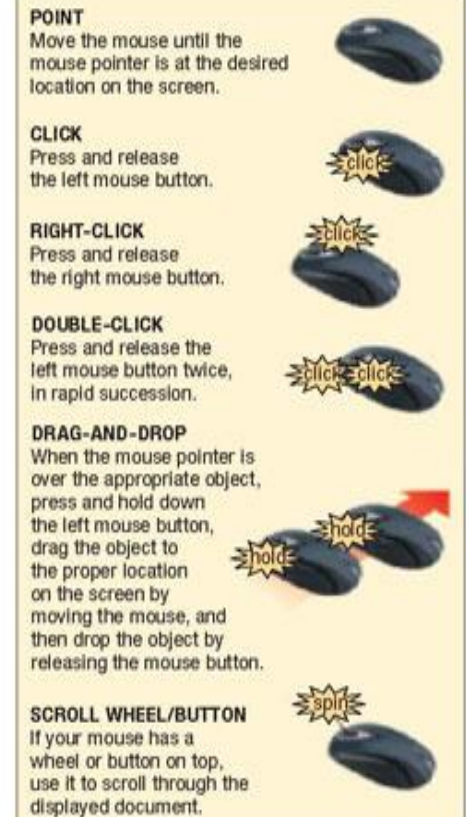


FIGURE 4-4
Common mouse operations.

Pens/Styluses

- Stylus: Pen-like device used to draw or write electronically on the screen
- Also called digital pen, electronic pen, tablet pen
- Commonly used with pen-based computers
 - Used to issue commands and input data
 - If handwriting recognition is used, written text can be converted to editable typed text



Handwriting Recognition

- Digital form: Used to input handwritten data into the computer and then convert it to editable text

FIGURE 4-6
Digital forms. If the software supports it, the text handwritten on a digital form can be converted by the computer to typed text.

The image displays two overlapping screenshots of a digital form titled "SKILLED NURSING VISIT NOTE". The form is part of the "Mi-Forms" software. The top screenshot shows the form with handwritten text in various fields, including "PATIENT NAME", "DATE", "TIME", and "NURSE". The bottom screenshot shows the same form with the handwritten text converted into typed, editable text. A large, semi-transparent watermark "Watermarkly" is visible across the center of the image.

Pens/Styluses

- Other uses for pens/styluses:
 - Digital writing systems
 - Graphics tablets
 - Signature capture devices



FIGURE 4-7

Other uses for digital pens.



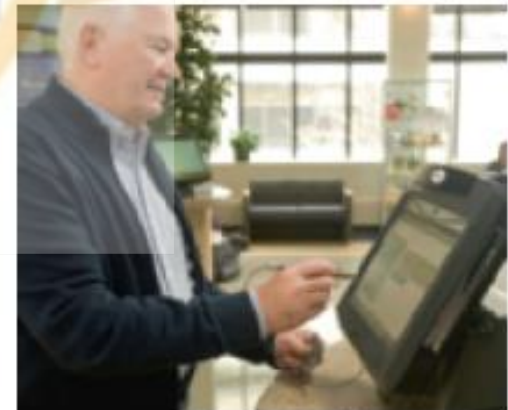
DIGITAL WRITING SYSTEMS

Record all input written on the paper and transfer it wirelessly to a computer upon demand.



GRAPHICS/PEN TABLETS

Transfer all input written or drawn on the tablet to the computer in real time and allow the use of pen flicks and other pen navigation tools.



SIGNATURE CAPTURE DEVICES

Record signatures for purchases, deliveries, and other applications that require recorded authorization.

Touch Screens

- Touch screen: Display device that is touched with the finger to select commands or otherwise provide input to the computer
- Used with:
 - Desktop and portable computers
 - Mobile phones and mobile devices
 - Surface computing
 - Consumer kiosks
- Can be multi-touch

Touch Screens

FIGURE 4-8
Touch screens.



DESKTOP COMPUTERS



MOBILE DEVICES



SURFACE COMPUTING DEVICES



CONSUMER KIOSKS

Other Pointing Devices

- Other pointing devices:
 - Joysticks, gamepads, and other gaming devices
 - Trackballs
 - Buttons and wheels
 - Touch pads



FIGURE 4-9

Other common pointing devices.

Quick Quiz

1. An optical mouse is _____.
 - a. the same as a wireless mouse
 - b. a mouse that tracks movements with light instead of a ball
 - c. a mouse that contains a scroll wheel on the top
2. True or False: With handwriting recognition, text is input as a graphical image so the text cannot later be edited as text.
3. An input device that looks like an upside-down mouse with the ball on top is a(n) _____.

Answers:

1) b; 2) False; 3) trackball

Scanners, Readers, and Digital Cameras

- Source documents: Documents containing data that already exists in physical form (order form, photograph, invoice, check, or price label)
- Source data automation: Capturing data directly from a source document
 - Saves time
 - Increases accuracy
 - Scanning or reading devices



RECORDING DATA DIRECTLY INTO A COMPUTER



CAPTURING DATA FROM ITS SOURCE DOCUMENT

FIGURE 4-10
Source data automation.
Recording data initially in digital form or capturing data directly from a source document can help reduce data input errors and save time.

Scanners

- Scanner (optical scanner): Input device that captures an image of an object and transfers them to a computer in digital form
 - Can scan photos, documents, drawings, (flat objects)
 - Data is typically input as a single image
 - If optical character recognition (OCR) is used, text is input as individual text characters
- Types of scanners
 - Flatbed
 - Portable
 - 3D
 - Integrated (ATMs, etc.)

Scanners

FIGURE 4-11
Optical scanners.



FLATBED SCANNERS

Used to input photos, sketches, slides, book pages, and other relatively flat documents into the computer.



PORTABLE SCANNERS

Used to capture small amounts of text; the text is typically transferred to a computer at a later time.



INTEGRATED SCANNERS

Built into other devices, such as into the ATM machine shown here to capture images of deposited checks.

Scanners

- Optical resolution: Quality of scanned images
 - Measured in number of dots per inch (dpi)
 - Can often be specified when image is scanned
 - Can be changed when scanned image is edited
 - Varies with scanner used
 - Higher resolution = better quality but larger file size



FIGURE 4-12

Scanning resolution.



RESOLUTION

Most scanners let you specify the resolution (in dpi) to use for the scan. High-resolution images look sharper but result in larger file sizes.

Barcode Readers

- Barcode readers: Input devices that read barcodes
- Barcode: Machine-readable code that represents data as a set of bars
 - Common types
 - Universal Product Code (UPC)
 - ISBN
 - Code 39
 - Intelligent mail code
 - 2D (QR) – hold more data



Barcode Readers

FIGURE 4-14
Barcode readers.



FIXED BARCODE READERS

Used most often in retail point-of-sale applications.



PORTABLE BARCODE READERS

Used when portability is needed.



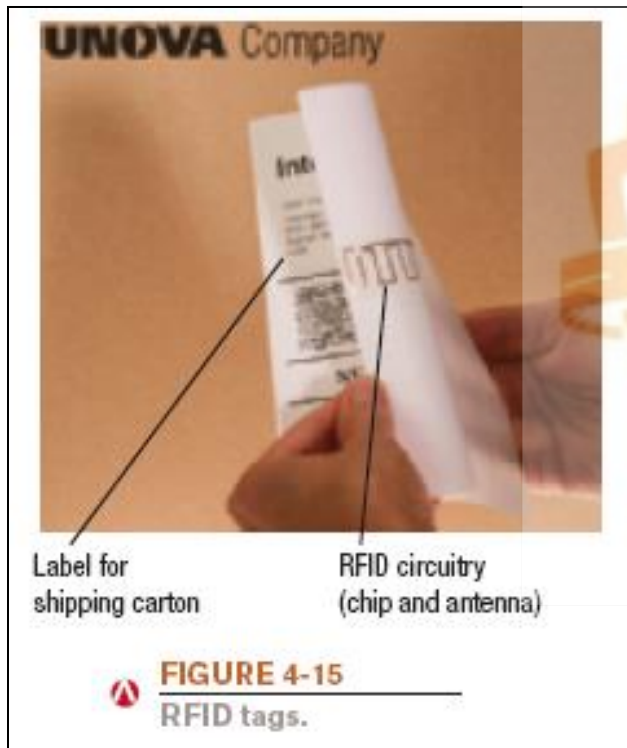
INTEGRATED BARCODE READERS

Used most often for consumer applications.

Radio Frequency Identification (RFID) Readers

- Radio frequency identification (RFID): Technology used to store and transmit data located in RFID tags
- RFID tag: Contains tiny chips and radio antennas
 - Attached to objects for identification purposes
 - Read by RFID readers
 - Tags only need to be within range of the reader, rather than in the line of sight
- Applications
 - Tracking inventory and assets
 - Electronic tolls
 - Tracking patients in hospitals
 - Ticketing applications
 - Security: Speeding up ID process

Radio Frequency Identification Readers



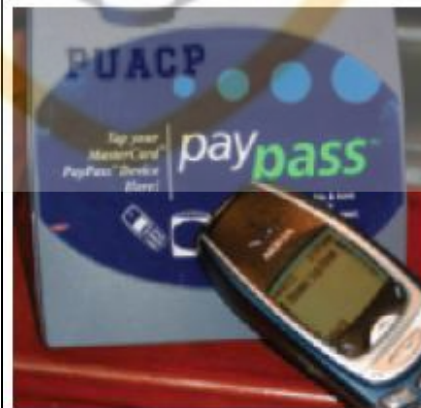
INVENTORY TRACKING

This portal RFID reader reads all of the RFID tags attached to all of the items on the pallet at one time.



TICKETING APPLICATIONS

This stationary RFID reader is used to automatically open ski lift entry gates for valid lift ticket holders at a ski resort in Utah.



MOBILE PAYMENTS

This stationary RFID reader is used at checkout locations to quickly process payments via RFID-enabled credit cards or mobile phones.



BORDER SECURITY

This stationary RFID reader is used at the U.S.-Mexico border crossing located in San Diego to reduce wait time.

FIGURE 4-16
RFID applications.

Optical Mark Readers (OMRs) and Optical Character Recognition (OCR) Devices

- Optical mark readers (OMRs): Input data from special forms to score or tally exams, questionnaires, ballots
- Optical character recognition (OCR): The ability of a computer to recognize scanned text characters and convert them to electronic form as text, not images
 - OCR readers can recognize many typed fonts
 - Used to process turnaround documents like monthly bills

FIGURE 4-18
Optical characters.
The most common use of optical characters is in turnaround documents, such as on the utility bill shown here.

PLEASE RETURN THIS PORTION WITH PAYMENT

NVEnergy.

Service Address: 123 MAPLE ST.
LAS VEGAS NV 89135

ACCOUNT NUMBER: 3000111111111111139

BALANCE FORWARD	.00
CURRENT CHARGES	135.86
TOTAL AMOUNT DUE	\$135.86

Current Charges due by Apr 5, 2011

Please enter amount paid below
\$ _____

9905.3.86.16458 T AV 0.124 sz 0.733

JOHN SMITH
123 MAPLE ST.
LAS VEGAS NV 89135

89520-3086

3000111111111111139 0000013586 0000013586 0 000

OPTICAL CHARACTERS
These OCR characters indicate the customer account number and amount due and can be read by both computers and humans.

Magnetic Ink Character Recognition (MICR) Readers

- Magnetic ink character recognition (MICR) readers:
Read MICR characters
 - Used primarily for banking
 - MICR readers read the special magnetic characters and sort/process checks
 - Can be used for remote deposit



Biometric Readers

- Biometric data: Based on unique physiological characteristics or personal trait
 - Fingerprint
 - Hand or face geometry
 - Iris of the eye
 - Voice or signature
- Biometric readers: Used to input biometric data
 - Can be stand-alone or built into another piece of hardware
 - Used to allow access only by authorized individuals
 - Most often used for access control and to verify transactions

Biometric Readers



STAND-ALONE FINGERPRINT READERS

Often used to control access to facilities or computer systems, such as to the notebook computer shown here.



BUILT-IN FINGERPRINT READERS

Typically used to control access to the device into which the reader is built, such as to the external hard drive shown here.



FIGURE 4-20

Biometric readers.

Digital Cameras

- Digital cameras: Record images on digital storage medium rather than film
 - Can either be still cameras or video cameras
 - Integrated into many portable computers and mobile phones.
- Digital still cameras
 - Available in a wide variety of sizes and capabilities
 - Primary appeal is images immediately available
 - Camera quality is measured in megapixels
 - Typically use flash memory for storage
 - Camera phones can be used to read barcodes, for mobile deposit, etc.

Digital Still Cameras



Digital Cameras

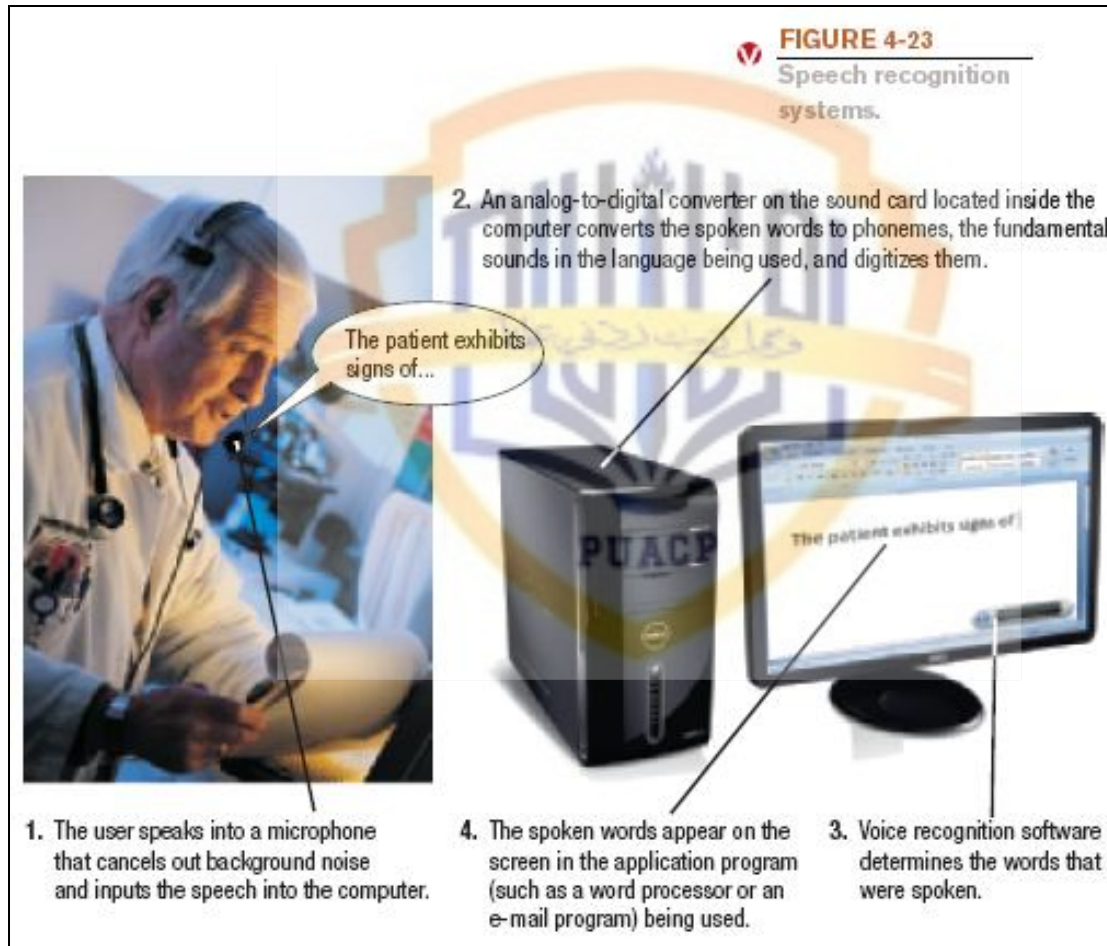
- Digital video cameras
 - Digital camcorders
 - PC video cameras (PC cams, Web cam)
 - Store images on digital media (flash memory, DVDs, hard drives, etc.)
- Applications:
 - Surveillance video cameras
 - Video conferences and Webinars
 - Face recognition systems



Audio Input

- Audio input: The process of entering audio data into the computer
- Voice input: Inputting spoken words and converting them to digital form
 - Via microphone or headset
 - Recorded for narrations, podcasts, etc.
 - VoIP (Voice over IP systems) applications
 - To provide spoken instructions to computer (speech recognition systems)
- Music input systems are used to input music
 - Microphones, keyboard controllers, etc.

Voice Input Systems



Quick Quiz

1. Which of the following is used in conjunction with Scantron test forms, voting ballots, and other documents in which the selection is bubbled in?
 - a. OCR
 - b. MICR
 - c. OMR
2. True or False: Flatbed scanners can be used to scan photos, as well as documents on conventional paper.
3. A voice input system requires software and a(n) _____ in order to input voice data or commands into a computer.

Answers:

1) c; 2) True; 3) microphone

Display Devices

- Display device: Presents output visually
 - Monitor: Display device for a desktop computer
 - Display screen: Screen built into a variety of devices
 - Notebook and other portable computers
 - Mobile phones and mobile devices
 - Handheld gaming devices, home entertainment devices, kitchen appliances
 - Digital photo frames, e-book readers
 - Digital signage systems, digital billboards

Display Devices



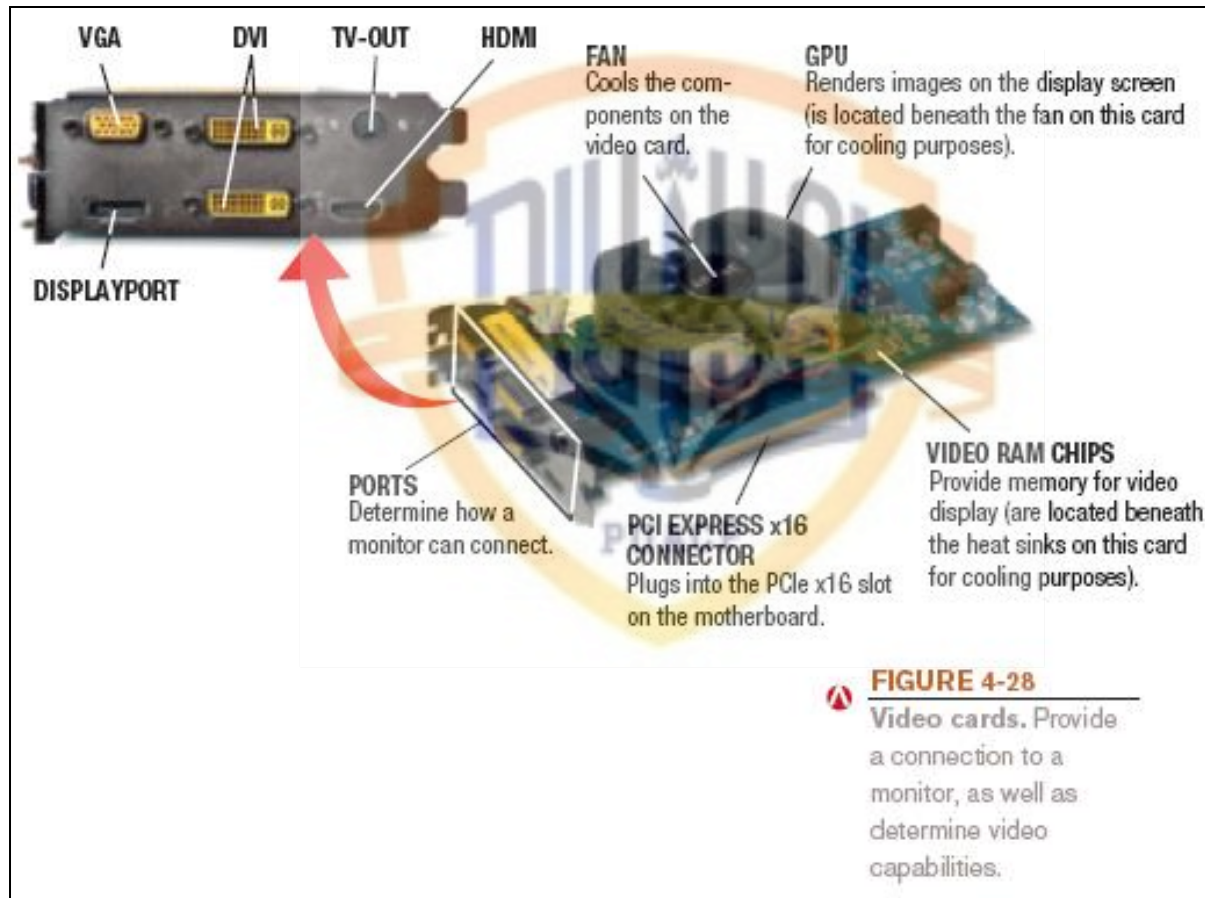
FIGURE 4-25
Uses for display devices.

Display Device Characteristics

- Color vs. monochrome
- CRT vs. flat-panel displays
- Size and aspect ratio
- Screen resolution
- Video adapters, interfaces, and ports
- Wired vs. wireless displays
- 2D vs. 3D
- Touch and gesture capabilities



Video Adapters



Flat-Panel Display Technologies

- Liquid crystal displays (LCDs): Use charged liquid crystals between sheets of glass or plastic
 - Requires backlighting
- LED (Light emitting diode): Used in displays as well as a variety of consumer products
- OLED (Organic Light emitting diode) – Uses layers of organic material
 - Emit visible light when current is applied
 - FOLED (Flexible OLED)
 - TOLED (Transparent OLED)
 - PHOLED (Phosphorescent OLED)

Flat-Panel Display Technologies

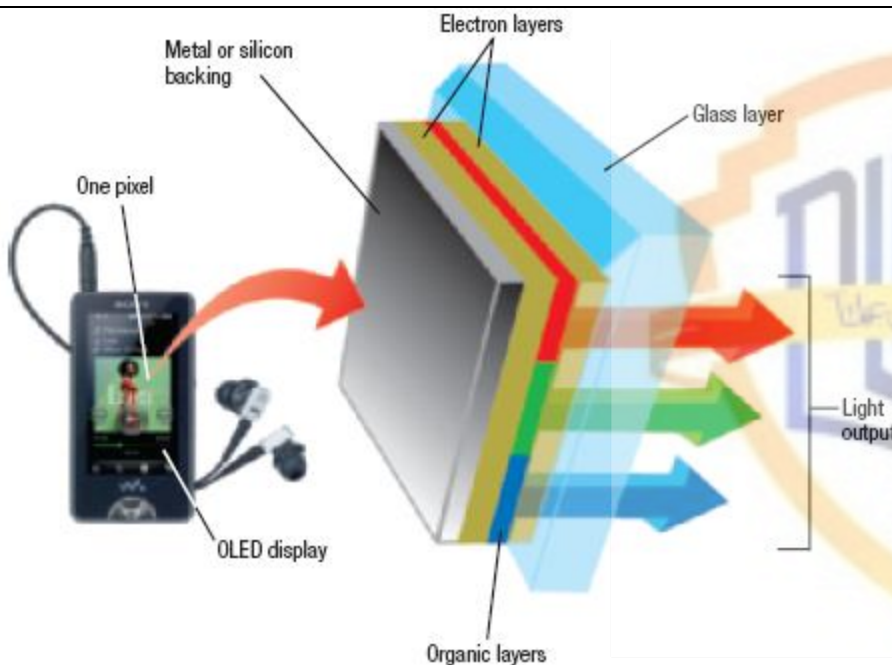


FIGURE 4-31
How OLED displays work. Each pixel on an OLED display emits light in the necessary color.



FOLEDs
Used to create flexible displays on plastic or another type of flexible material.



TOLEDs
Used to create transparent displays.

FIGURE 4-32
Special types of OLEDs.

Flat-Panel Display Technologies

- Interferometric modulator displays (IMOD): Essentially a complex mirror that uses external light to display images
 - Designed initially for mobile phones and portable devices
 - Images are bright and clear, even in sunlight
- Plasma displays: Use layers of gas to display images
 - Most often used on large displays
- Surface-conduction electron-emitter displays (SED): Millions of tiny electron guns, similar to CRT
 - Thin, bright, less flicker than LCD or plasma

Data and Multimedia Projectors

- Data projector: Display device that projects all computer output to a wall or projection screen
 - Found in classrooms, conference rooms
 - Can be wireless or integrated into devices
 - Integrated – tiny projectors built into mobile phones, portable devices
 - Hologram projectors used to display 3D images

FIGURE 4-34
Data projectors.



CONVENTIONAL DATA PROJECTORS
Frequently used for both business and classroom presentations.



INTEGRATED AND PORTABLE PROJECTORS
Images displayed on the device (such as the mobile phone shown here) are projected onto any surface.



HOLOGRAPHIC PROJECTORS
Project 3D images of objects or individuals.

Printers

- Printers: Produce hard copy
- Printer characteristics
 - Printing technology used
 - Impact vs. nonimpact
 - Color vs. black and white
 - Personal vs. network printers
 - Print resolution (measured in dpi)
 - Print speed (measured in pages per minute (PPM))
 - Connection (USB, Ethernet, Wi-Fi, etc.)
 - Multifunction capabilities

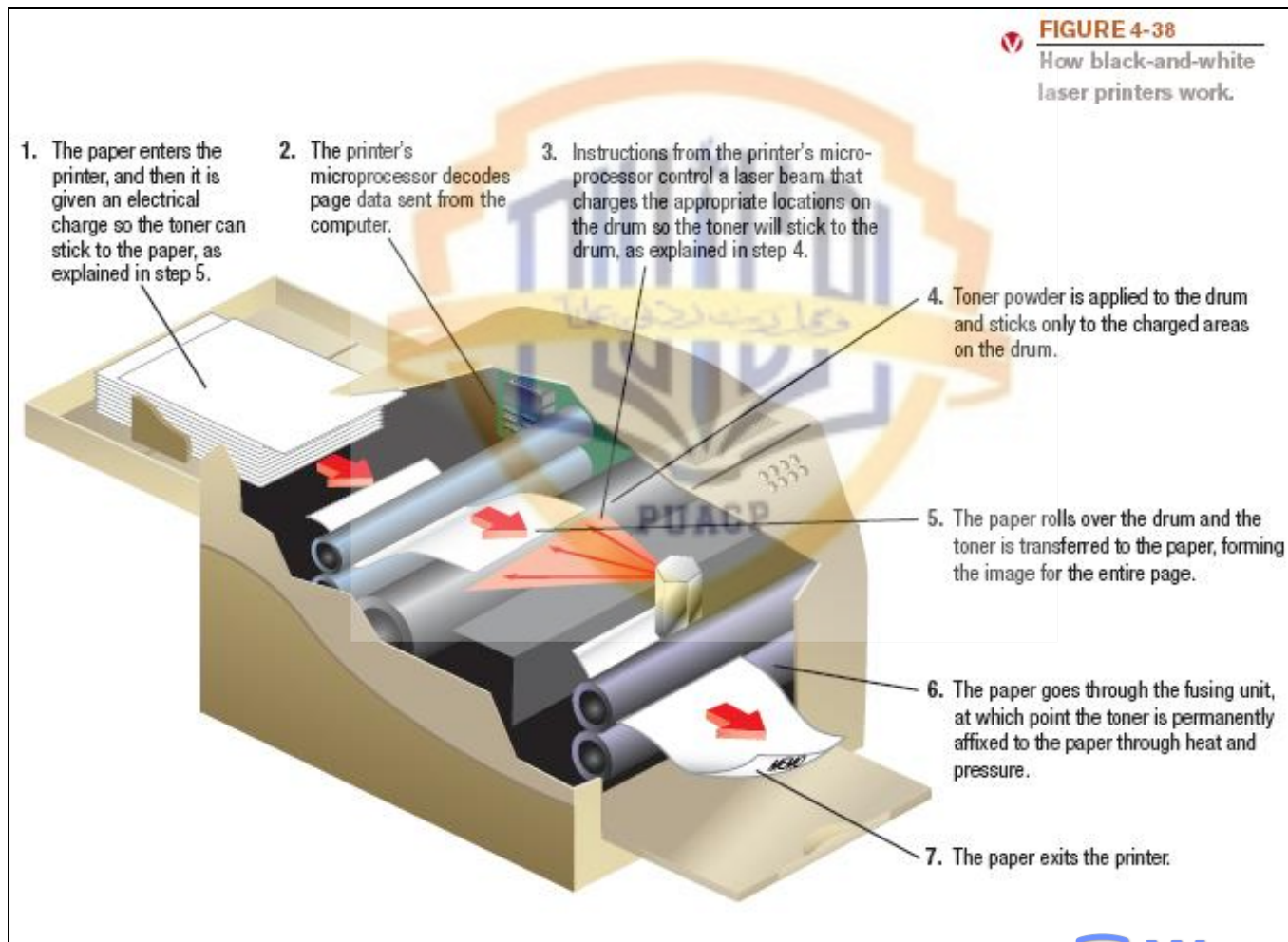


FIGURE 4-35
▲ **Dot-matrix printers.** Dot-matrix printers are impact printers; today they are typically high-speed printers used in manufacturing, shipping, or similar applications.

Laser Printers

- Laser printer: Uses toner powder and technology similar to that of a photocopier to produce images on paper
 - The standard for business documents
 - Print one entire page at a time
 - Generally faster and have better quality than ink-jet printers
 - Can be black and white or color
 - Common print resolution for laser printers is between 600 and 2,400 dpi
 - Use toner cartridges

Laser Printers



Ink-Jet Printers

- Ink-jet printer: Sprays droplets of ink to produce images on paper
 - Usually print in color
 - Often the choice for home use
 - Print fairly slowly, one line at a time
 - Quality not quite as good as a laser printer
 - Use ink-jet cartridges
- Newer printers with full width printheads are much faster
- Potential applications for the future
 - Dispensing liquid metal, aromas, computer chips and other circuitry, “printing” human tissue

Ink-Jet Printers

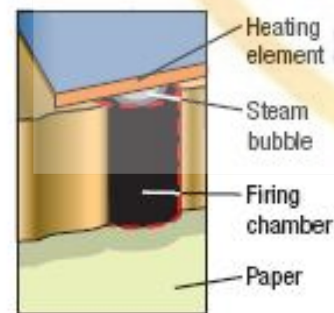
FIGURE 4-39

How ink-jet printers work.



Each ink cartridge is made up of multiple tiny ink-filled firing chambers; to print images, the appropriate color ink is ejected through the appropriate firing chamber.

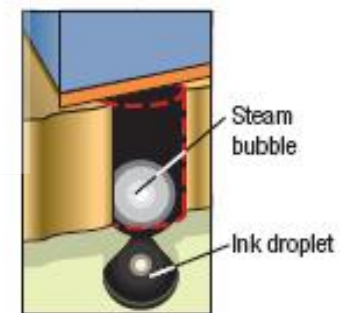
Ink-jet printer



1. A heating element makes the ink boil, which causes a steam bubble to form.



2. As the steam bubble expands, it pushes ink through the firing chamber.



3. The ink droplet is ejected onto the paper and the steam bubble collapses, pulling more ink into the firing chamber.

Special-Purpose Printers

- Photo printers
- Barcode, label, and postage printers
- Portable printers
- Plotters and wide-format ink-jet printers
- 3-D printers



Audio Output

- Audio output: Output in the form of voice, music, and other audible sounds
 - Speakers
 - Headphones and headsets
 - Earphones and earbuds

FIGURE 4-41
Audio output devices.



COMPUTER SPEAKERS

Used to output sound from a computer.



IPOD/MP3 DOCK

Used to output sound from a portable digital media player.



HEADSETS

Used when both voice input and audio output are required.

Quick Quiz

1. Which of the following types of display devices should have the largest footprint (the amount of room taken up on a desk)?
 - a. CRT monitor
 - b. OLED display
 - c. LCD display
2. True or False: Laser printers can only print in black and white.
3. _____ printers form images with drops of liquid ink.

Answers:

1) a; 2) False; 3) Ink-jet printers

Summary

- Keyboards
- Pointing and Touch Devices
- Scanners, Readers, and Digital Cameras
- Audio Input
- Display Devices
- Printers
- Audio Output

